

Pre-work • Meninges, CSF, and the Spinal Cord

The Meninges and Cerebrospinal Fluid, and the Spinal Cord. Read both Part A chapters first. These two chapters are layers and routes rather than lists. Work them with your finger tracing the order, and the naming comes with it.

Module 5 • 3 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first.
This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the meninges and cord chapters into mini visuals. Eight chunks. The ladders carry the layers, the chain carries the fluid, and the hubs carry the two structures you will be asked to draw from memory.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 LADDER

The meninges, outside in

Rungs from the skull down to the brain surface: periosteal dura, meningeal dura, arachnoid, pia. Write the space between each pair of rungs on the line between them, and mark the one space that holds cerebrospinal fluid. Then note beside the top rung why the cranium has no epidural space and the vertebral canal does.

2 GRID

The three meningeal spaces

Space down the side. Location, contents, and one clinical use or bleed across the top. Add a fourth column: present in the cranium, yes or no. The subdural row is the one people get wrong, so write potential space in it and underline it.

3 HUB

The dural reflections and the sinuses they carry

A brain in the cranial cavity at the hub. Draw the falx cerebri dipping into the longitudinal fissure, the tentorium cerebelli between cerebrum and cerebellum, and the falx cerebelli below it. Then draw the superior sagittal sinus in the top edge of the falx cerebri and follow it back to the confluence, the transverse sinuses, the sigmoid sinuses, and the internal jugular veins.

4 GRID

The four cerebrospinal fluid spaces

Ventricle down the side, lateral through to the central canal. Location, what it connects to, and the name of the opening it connects through across the top. Add a fourth column: does it hold choroid plexus. Three rows say yes and one says no.

DRAWING 1

The meninges over the top of the brain, in coronal section

BRAIN DUMP FIRST

Two minutes. Every layer from bone to brain, and every space between them.

Draw a wedge of skull with the brain beneath it, cut across the midline. Lay in the periosteal dura against the bone and the meningeal dura beneath it, and show the two layers parting at the midline to enclose the superior sagittal sinus. Draw the falx cerebri dipping down into the longitudinal fissure between the hemispheres. Add the arachnoid mater with arachnoid granulations poking up into the sinus, the subarachnoid space filled with CSF, and the pia mater following every gyrus and sulcus.



In your notebook, head the page **M5-1 The meninges over the top of the brain, in coronal section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name every layer your needle would cross going from the scalp to the cortex, and say which of them a cranial epidural bleed sits outside of.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A clinician places a needle in the midline between the L4 and L5 spinous processes and withdraws clear fluid. Name the pool the needle reached and explain why the spinal cord is not injured.

Answer. The lumbar cistern, the enlarged subarachnoid space below the conus medullaris.

Reasoning. The answer comes from a mismatch in growth, not from the needle. The spinal cord stops lengthening at about four to five years of age while the vertebral column keeps growing, so in the adult the cord ends at the conus medullaris near the L1 to L2 disc, well above the entry point. The meninges do not stop there: the subarachnoid space runs on down as the lumbar cistern, so there is still cerebrospinal fluid to draw. What occupies that canal below the conus is the cauda equina, a loose bundle of nerve roots that floats in the fluid and drifts aside from a blunt needle. An entry above L2 would be a different question entirely, because the cord itself would be in the way, and that is why the level is part of the answer.

Set A - Name it

1. The meningeal layer that clings directly to the surface of the brain is the _____.
2. The dural fold separating the cerebrum from the cerebellum is the _____.
3. The network of capillaries wrapped in ependymal cells that produces CSF is the _____.
4. The tapered inferior end of the spinal cord is the _____.
5. Name the three components of the blood-brain barrier.

5 CHAIN

The circulation of CSF, made to reabsorbed

Eight boxes with arrows, lateral ventricles through to the arachnoid granulations. Mark every narrow point on the route with a circle: the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. Those three circles are where a blockage produces hydrocephalus.

6 LADDER

The layers a lumbar puncture crosses

Nine rungs, skin at the top and the subarachnoid space at the bottom. Write the three ligaments in the correct order in the middle, and mark the rung where the clinician feels the resistance give way. Beside the ladder write the vertebral level the needle enters and the level the cord ends at.

7 HUB

The spinal cord in cross section

The central canal at the hub. Draw the gray butterfly around it with the posterior, lateral, and anterior horns, and write what each horn holds. Then draw the white matter around the gray as anterior, posterior, and lateral columns. Add the rootlets, both roots, the posterior root ganglion, and the spinal nerve leaving on one side only.

8 GRID

Telling the cord segments apart

Segment down the side, cervical through coccygeal. Diameter, amount of white matter, amount of gray matter, and shape across the top. Then add one column for the lateral horn, present or absent, because only two segments have it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

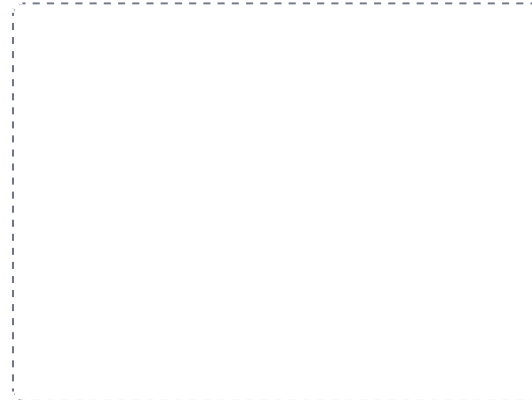
DRAWING 2

The ventricles and the whole path of the CSF

BRAIN DUMP FIRST

Two minutes. Four spaces, three narrow points, and where the fluid leaves the system.

Draw the brain in outline from the side and put the ventricles inside it: a lateral ventricle curving through the hemisphere, the third ventricle in the diencephalon, the cerebral aqueduct through the midbrain, and the fourth ventricle between the brainstem and the cerebellum, continuing into the central canal. Mark the choroid plexus in each ventricle that has it. Then arrow the fluid out through the apertures into the subarachnoid space, over the surface, and into the superior sagittal sinus at the arachnoid granulations.



In your notebook, head the page **M5-2 The ventricles and the whole path of the CSF** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point to any one narrow spot on your drawing and say which ventricles would enlarge if it were blocked.

Set B · Predict it

1. The cerebral aqueduct is narrowed by a tumor. Predict which ventricles enlarge and which do not, and name the condition.

2. The arachnoid granulations are scarred and stop working. Predict what happens to CSF volume and pressure, and say why production does not simply stop.

3. A cross-section of cord shows a small lateral horn. Predict which region of the cord it came from and give your reason.

4. A cord injury is sustained at the T6 level. Predict which functions are lost and which are spared, using the tracts.

Set C · Tell it apart

1. Distinguish an epidural hematoma from a subdural hematoma from a subarachnoid hemorrhage by the space each one fills.

2. Explain why the vertebral canal has a true epidural space and the cranium does not, using the layers of the dura.

3. Compare a cervical cord section with a sacral cord section by the proportion of gray to white matter, and explain the reason for the difference.

DRAWING 3

The spinal cord, whole and in cross section

BRAIN DUMP FIRST

Two minutes. The outside of the cord, where it ends, and what a slice through it shows.

Draw the whole cord from the medulla down, inside the vertebral canal. Mark the cervical enlargement and the lumbar enlargement, taper the cord into the conus medullaris at the L1 to L2 disc, run the filum terminale down to the coccyx, and hang the cauda equina around it. Then draw one cross-section beside it: anterior median fissure, posterior median sulcus, gray horns, white columns, central canal, both roots, and the posterior root ganglion.



In your notebook, head the page **M5-3 The spinal cord, whole and in cross section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your own drawing why a needle placed between L4 and L5 draws fluid without touching the spinal cord.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

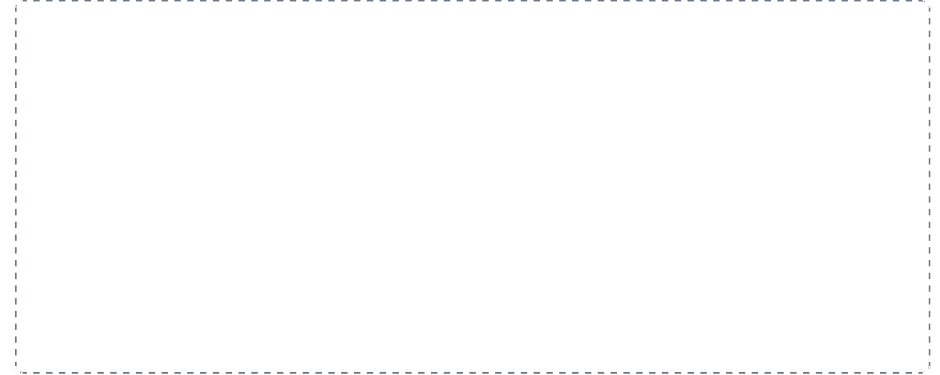
1 The meninges, outside in

LADDER



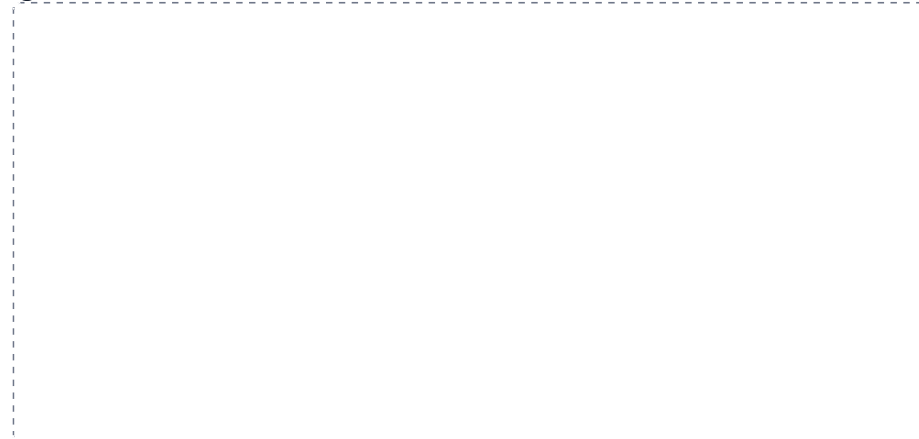
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GRID



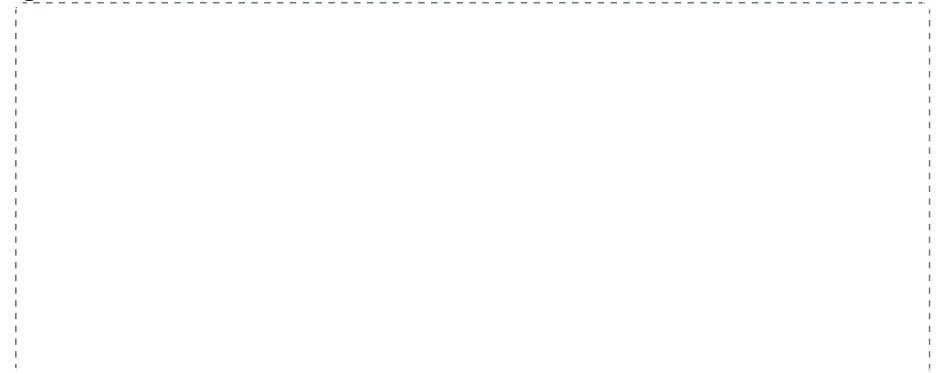
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4 The four cerebrospinal fluid spaces

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5 The circulation of CSF, made to reabsorbed

CHAIN

6 The layers a lumbar puncture crosses

LADDER

7 The spinal cord in cross section

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8 Telling the cord segments apart

GRID